

- Description

This module manages a smart water distribution network pressure monitoring system, ensuring that:

1. **The pressure in water pipelines is continuously measured and maintained within safe limits.**
2. **If the pressure exceeds the maximum threshold, an overpressure warning is triggered.**
3. **If the pressure drops below the minimum threshold, a leak detection alert is issued.**
4. **A log of all pressure readings, alerts, and maintenance actions is maintained for system efficiency.**

This system helps prevent pipe bursts, detect leaks early, and optimize water supply management.

- Specification

module SmartWaterPipelineMonitoring

var currentPressure, maxPressure, minPressure: real

var log: sequence of string

process updatePressure(changeAmount: real)

functional scenario form:

currentPressure $\geq$ minPressure && log = log $\oplus$ ("Water Pressure Updated: " + currentPressure + " bar"), Water pressure updated successfully.

||

currentPressure > maxPressure && log = log $\oplus$ ("Overpressure Warning: " + currentPressure + " bar, Risk of Pipe Bursting."), Overpressure warning triggered, potential pipe damage.

||

currentPressure < minPressure && log = log $\oplus$ ("Leakage Alert: " + currentPressure + " bar, Immediate Inspection Required."), Leak detection alert triggered, inspection required.

end process

end module

- Java Code

```
package code;
import java.util.*;
class WaterPipeline {
    private double maxPressure;
    private double minPressure;
    private double currentPressure;
    private List<String> log;
    public WaterPipeline(double maxPressure, double minPressure, double
initialPressure) {
```

```

        this.maxPressure = maxPressure;
        this.minPressure = minPressure;
        this.currentPressure = initialPressure;
        this.log = new ArrayList<>();
    }
    public void updatePressure(double changeAmount) {
        currentPressure += changeAmount;
        String message = String.format("Current Water Pressure: %.2f bar", currentPressure);
        if (currentPressure > maxPressure) {
            message += " | WARNING: Overpressure Detected! Risk of Pipe Bursting.";
        } else if (currentPressure < minPressure) {
            message += " | ALERT: Possible Leak Detected! Immediate Inspection Required.";
        }
        System.out.println(message);
        log.add(message);
    }
    public void printLog() {
        System.out.println("\nWater Pressure Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c41_WaterDistributionSystem {
    public static void main(String[] args) {
        WaterPipeline pipeline = new WaterPipeline(8.0, 3.0, 5.0);
        // testcase-VT:
        //     pipeline.updatePressure(2.5);
        //     pipeline.updatePressure(-3.0);
        //     pipeline.updatePressure(-2.0);
        //     pipeline.updatePressure(4.0);
        //     pipeline.updatePressure(3.5);

        // testcase-FT:
        //     pipeline.updatePressure(-1.7976931348623157E308);
        //     pipeline.updatePressure(1.1444946573542756E297);
        //     pipeline.updatePressure(8.174961810401071E296);
        //     pipeline.updatePressure(0.0);
        //     pipeline.updatePressure(1.5368928377020603E298);

        // testcase-Z3:

```

```
//      pipeline.updatePressure(2);
//      pipeline.updatePressure(3);
//      pipeline.updatePressure(2);
//      pipeline.updatePressure(4);
//      pipeline.updatePressure(3);

// testcase-VT:
pipeline.updatePressure(2.5);
pipeline.updatePressure(-3.0);
pipeline.updatePressure(-2.0);
pipeline.updatePressure(4.0);
pipeline.updatePressure(3.5);

//      pipeline.printLog();
}
}
```

● Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	836	842
✓ src	0.7 %	6	836	842
✓ code	0.7 %	6	836	842
✓ c41_WaterDistributionSystem.java	100.0 %	6	0	6
> c41_WaterDistributionSystem		0	0	0
> WaterPipeline	100.0 %	6	0	6

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	106	20,598	20,704
✓ src	0.5 %	106	20,598	20,704
✓ code	0.5 %	106	20,598	20,704
✓ c41_WaterDistributionSystem.java	97.2 %	106	3	109
> c41_WaterDistributionSystem	89.3 %	25	3	28
> WaterPipeline	100.0 %	81	0	81

4. Fuzz testing branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c41_WaterDistributionSystem.java	66.7 %	4	2	6
> WaterPipeline	66.7 %	4	2	6
> c41_WaterDistributionSystem		0	0	0

5. Fuzz testing instruction coverage: 93.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	102	20,602	20,704
✓ src	0.5 %	102	20,602	20,704
✓ code	0.5 %	102	20,602	20,704
✓ c41_WaterDistributionSystem.java	93.6 %	102	7	109
> WaterPipeline	95.1 %	77	4	81
> c41_WaterDistributionSystem	89.3 %	25	3	28

6. Z3 branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	837	840
▼ src	0.4 %	3	837	840
▼ code	0.4 %	3	837	840
▼ c41_WaterDistributionSystem.java	50.0 %	3	3	6
> WaterPipeline	50.0 %	3	3	6
> c41_WaterDistributionSystem		0	0	0

7. Z3 instruction coverage: 76.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	82	22,298	22,380
▼ src	0.4 %	82	22,298	22,380
▼ code	0.4 %	82	22,298	22,380
▼ c41_WaterDistributionSystem.java	76.6 %	82	25	107
> WaterPipeline	72.8 %	59	22	81
> c41_WaterDistributionSystem	88.5 %	23	3	26

8. Unimproved V-method branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c41_WaterDistributionSystem.java	66.7 %	4	2	6
> WaterPipeline	66.7 %	4	2	6
> c41_WaterDistributionSystem		0	0	0

9. Unimproved V-method instruction coverage: 79.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	85	22,295	22,380
▼ src	0.4 %	85	22,295	22,380
▼ code	0.4 %	85	22,295	22,380
▼ c41_WaterDistributionSystem.java	79.4 %	85	22	107
> WaterPipeline	76.5 %	62	19	81
> c41_WaterDistributionSystem	88.5 %	23	3	26